

17	resistors in series	$R_T = R_1 + R_2$
18	resistors in parallel	$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$
19	capacitors	time constant : $\tau = RC$
20	Lorentz factor	$\gamma = \frac{1}{\sqrt{1 - v^2 / c^2}}$
21	time dilation	$t = t_o \gamma$
22	length contraction	$L = L_o / \gamma$
23	relativistic mass	$m = m_o \gamma$
24	total energy	$E_{total} = E_k + E_{rest} = mc^2$
25	universal gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
26	mass of Earth	$M_E = 5.98 \times 10^{24} \text{ kg}$
27	radius of Earth	$R_E = 6.37 \times 10^6 \text{ m}$
28	mass of the electron	$m_e = 9.1 \times 10^{-31} \text{ kg}$
29	charge on the electron	$e = -1.6 \times 10^{-19} \text{ C}$
30	speed of light	$c = 3.0 \times 10^8 \text{ m s}^{-1}$

Prefixes/Unitsp = pico = 10^{-12} n = nano = 10^{-9} μ = micro = 10^{-6} m = milli = 10^{-3} k = kilo = 10^3 M = mega = 10^6 G = giga = 10^9 t = tonne = 10^3 kg **END OF FORMULA SHEET**